

Covered: Submodules, Torsion, module homomorphisms, isomorphisms, direct sums, cyclic submodules, generating sets, linear (in)dependence

1 Modules (Continued.)

Let M be an R -module. For an ideal $I \subseteq R$, define

$$IM := \left\{ \sum_{i=1}^n r_i m_i \mid r_i \in I, m_i \in M, n \in \mathbb{N} \right\}.$$

Then IM is a submodule of M .

Recall: $N \subseteq M$ is a submodule iff

$$x + y \in N, \quad rx \in N \quad \forall x, y \in N, r \in R.$$

Definition 1.1. Let M be an R -module. Then $m \in M$ is called a *torsion element* if $rm = 0$ for some $0 \neq r \in R$.

$$\text{Tor}(M) := \{m \in M \mid \exists 0 \neq r \in R \text{ s.t. } rm = 0\}.$$

Proposition 1.2. Let R be an integral domain. Then $\text{Tor}(M)$ is a submodule of M .

Proof. Let $x, y \in \text{Tor}(M)$. Then

$$\exists r_1, r_2 \neq 0 \quad \text{s.t.} \quad r_1 x = r_2 y = 0.$$

Hence

$$r_1 r_2 (x + y) = r_2 (r_1 x) + r_1 (r_2 y) = 0,$$

and since R is an integral domain, $r_1 r_2 \neq 0$. Thus $x + y \in \text{Tor}(M)$. Also, for $r \in R$,

$$r_1 (rx) = r(r_1 x) = 0,$$

so $rx \in \text{Tor}(M)$. □

Let F be a field and $R = F[x]$. Let V be a vector space over F , and $T : V \rightarrow V$ a linear transformation. Then V is an R -module by

$$f(x) \cdot v := f(T)v, \quad f(x) \in R, v \in V.$$

In particular,

$$0 \cdot v = 0, \quad 1 \cdot v = v.$$

Let N be a submodule of M . Define

$$\text{ann}(N) := \{r \in R \mid rn = 0 \forall n \in N\}.$$

Question. Show that $\text{ann}(N)$ is an ideal of R .

Recall: A is an R -algebra if $\exists \varphi : R \rightarrow A$ a homomorphism s.t.

$$\varphi(1_R) = 1_A, \quad \varphi(R) \subseteq Z(A).$$

Given R -algebras A and B , an R -algebra homomorphism is a homomorphism $\psi : A \rightarrow B$ s.t.

$$\psi(1_A) = 1_B, \quad \psi \circ \varphi_A = \varphi_B.$$

Example 1.3. 1. Any commutative ring with 1 is a $\mathbb{Z}$ -algebra.

2. Let A be a commutative ring with identity. For any subring $R \subseteq A$ with $1_A \in R$, A is an R -algebra.

Definition 1.4. Let M, N be R -modules. An R -module homomorphism is a map $\varphi : M \rightarrow N$ satisfying

$$\varphi(x + y) = \varphi(x) + \varphi(y) \quad \forall x, y \in M,$$

and

$$\varphi(rx) = r\varphi(x) \quad \forall r \in R, x \in M.$$

Equivalently,

$$\varphi(rx + y) = r\varphi(x) + \varphi(y).$$

Also called a linear map or linear transformation.

Example 1.5.

$$T : \mathbb{C} \rightarrow \mathbb{C}, \quad R = \mathbb{R}, \quad z \mapsto \bar{z}.$$

If this map is bijective, then we call it an R -module isomorphism and write

$$M \xrightarrow[\varphi]{\cong} N \quad \text{or} \quad M \cong_R N.$$

Given $\varphi : M \rightarrow N$, define

$$\ker(\varphi) := \{m \in M \mid \varphi(m) = 0\},$$

and

$$\text{im}(\varphi) := \{\varphi(m) \mid m \in M\} = \{n \in N \mid \exists m \in M \text{ with } \varphi(m) = n\}.$$

Let $\varphi, \psi : M \rightarrow N$. Then $\varphi + \psi$ and, for $r \in R$, $r\varphi$, where

$$(r\varphi)(m) := r\varphi(m),$$

are also R -homomorphisms.

Define

$$\text{Hom}_R(M, N) := \{\varphi : M \rightarrow N \mid \varphi \text{ is an } R\text{-homomorphism}\}.$$

Example 1.6. 1. Given R , let $M = R^n$ and $N = R$. Define

$$\pi_i : R^n \rightarrow R, \quad \pi_i(x_1, \dots, x_n) = x_i.$$

2. Let $M = N = R$. Then

$$\text{Hom}_R(R, R) \cong R.$$

Given $\varphi : R \rightarrow R$,

$$\varphi(r) = \varphi(r \cdot 1) = r\varphi(1).$$

Given $a \in R$, $r \mapsto ra$ is also an R -homomorphism.

3.

4. $M = \mathbb{Z}[i]$ and $N = \mathbb{Z}[\sqrt{2}]$. As rings,

$$M \not\cong N,$$

but as $\mathbb{Z}$ -modules,

$$M \cong_{\mathbb{Z}} N.$$

Indeed,

$$M = \mathbb{Z} \oplus i\mathbb{Z} \cong \mathbb{Z}^2, \quad N = \mathbb{Z} \oplus \sqrt{2}\mathbb{Z} \cong \mathbb{Z}^2.$$

Proposition 1.7. $\text{Hom}_R(M, M)$ is a ring with 1, and it is an R -algebra.

Definition 1.8. We call $\text{Hom}_R(M, M)$ the *endomorphism ring* of M and it is denoted by

$$\text{End}_R(M).$$

Observe: Let M be an R -module and N be a submodule. Then M/N is an R -module where

$$r(m + N) = rm + N \quad \forall r \in R, m \in M.$$

Theorem 1.9 (Isomorphism Theorems for Modules). 1. $\varphi : M \rightarrow N$, R -homomorphism. Then

$$M/\ker \varphi \cong \varphi(M).$$

2. A, B submodules of R -module M . Then

$$(A + B)/B \cong A/(A \cap B).$$

3. Let $A \subseteq B$ be submodules. Then

$$(M/A)/(B/A) \cong M/B.$$

4. N submodule of M . Then

$$\{L : N \subseteq L \subseteq M\} \longleftrightarrow \{\text{submodules of } M/N\}, \quad L \mapsto L/N \quad \forall L \supseteq N.$$

Definition 1.10. Given R -modules M and N , define the direct sum as

$$M \oplus N := \{(m, n) \mid m \in M, n \in N\}.$$

Then $M \oplus N$ is an R -module with

$$\begin{aligned} (m_1, n_1) + (m_2, n_2) &= (m_1 + m_2, n_1 + n_2), \\ r(m, n) &:= (rm, rn). \end{aligned}$$

Definition 1.11. Let M be an R -module and $N_1, N_2, \dots, N_m$ be submodules. The sum

$$N_1 + \dots + N_m := \{v_1 + \dots + v_m \mid v_i \in N_i\}.$$

Given a collection $\{M_i\}_{i \in I}$ of R -modules, define

$$\bigoplus_{i \in I} M_i := \{(m_i)_{i \in I} \mid m_i \in M_i \text{ and } m_i \neq 0 \text{ for only finitely many } i\}.$$

$$\text{Direct product} = \prod_{i \in I} M_i := \{(m_i)_{i \in I} \mid m_i \in M_i\}.$$

Example 1.12.

$$M \cong M \oplus \{0\} \quad \text{and} \quad N \cong \{0\} \oplus N,$$

so that M, N are submodules of $M \oplus N$.

Definition 1.13. Let M be an R -module and $N_1, N_2, \dots, N_m$ be submodules. The sum

$$N_1 + \dots + N_m = \{n_1 + \dots + n_m \mid n_i \in N_i\}.$$

Given any subset $A \subseteq M$, define

$$RA := \{r_1 a_1 + \dots + r_n a_n \mid r_i \in R, a_i \in A, n \in \mathbb{N}\}.$$

If $A = \emptyset$, then $RA = \{0\}$ and if $A = \{0\}$, then $RA = \{0\}$. If $A = \{a_1, \dots, a_n\}$, then $RA = Ra_1 + \dots + Ra_n$.

Definition 1.14. If N is a submodule such that $N = RA$ we say N is *generated by* A . If A is finite, then N is *finitely generated*.

Definition 1.15. N is cyclic if

$$\exists a \in M \quad \text{s.t.} \quad N = Ra.$$

Given a finitely generated submodule N , let

$$d := \min\{|A| : N = RA\}.$$

Then any set A with

$$|A| = d, \quad N = RA$$

is called a *minimal generating set* of N .

Example 1.16.

$$\begin{aligned} \mathbb{Z}[\sqrt{2}] &= \mathbb{Z}A, & A &= \{1, \sqrt{2}\} \\ \mathbb{Z} &= \mathbb{Z}A_1, & A_1 &= \{2, 3\} \\ \mathbb{Z} &= \mathbb{Z}A_2, & A_2 &= \{1\} \end{aligned}$$

Exercise 1.17. Dummit and Foote: Pages 343-343 : 2, 6, 7, 9-14, 21. Page 350: 5, 6, 8, 10, 13

A set $A \subseteq N$ is a generating set for N iff every $n \in N$ can be written in the form

$$n = \sum_{i=1}^k r_i a_i, \quad r_i \in R, \quad a_i \in A, \quad k \in \mathbb{N}.$$

Definition 1.18. Let $\{m_i\}_{i \in I} \subseteq M$. We say $\{m_i\}_{i \in I}$ is *linearly independent* if

$$\sum_{i \in I} r_i m_i = 0 \implies r_i = 0 \quad \text{for all } i \in I,$$

where only finitely many r_i 's are nonzero. Otherwise, it is *linearly dependent*.

We call A a *basis* for M if $M = RA$ and A is linearly independent. A module which has a basis is called a *free module*. If the basis is finite, then it is a *finite free module*.

Example 1.19. 1. Finite-dimensional vector spaces are finite free modules.

2. If R is a commutative ring with unity, then $R[x]$ is a free R -module: $\{1, x, x^2, \dots\}$ is a basis.

3.

4. Let

$$R = \mathbb{Z}[\sqrt{-5}], \quad I = (2, 1 + \sqrt{-5}).$$

Then R is both an R -module and a $\mathbb{Z}$ -module. Moreover,

$$\{1, 2 + \sqrt{-5}\}$$

is linearly dependent over R , but linearly independent over $\mathbb{Z}$.

Remark 1.20. $\{2, 3\}$ are linearly dependent over $\mathbb{Z}$, but we cannot write one as a multiple of the other.

So, if $\{a_1, \dots, a_m\}$ are linearly dependent, we may not be able to write a_i as a linear combination of the remaining a_j 's.

Remark 1.21. A finitely generated module need not have a basis. For example,

$$R = \mathbb{Z}[\sqrt{-5}], \quad I = (2, 1 + \sqrt{-5}) = 2R + (1 + \sqrt{-5})R.$$

Exercise 1.22. Given $x, y \in \mathbb{Z}$, show $\exists a, b \in R$ such that $ax + by = 0$ and $(a, b) \neq (0, 0)$.